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{
  "creator" : "ari.neko@cvut.cz",
  "timestamp" : "09-07-2026_17:23:51",
  "variant" : 3,
  "subjectName" : "java-programming",
  "categories" : [ "Generics & Types", "Exception Handling", "Multi-threading & Concurrency" ],
  "questions" : [ {
    "question" : "What does the wildcard bound pattern 'List<? extends Number>' accomplish inside a method signature?",
    "answers" : [ "It turns the List into an immutable collection", "It opens the List to accept additions of any custom object structure", "It mandates that all elements in the List must carry matching static values", "It restricts the parameter to a List containing Number or any of its subclasses" ],
    "correctIndex" : 3
  }, {
    "question" : "What primary objective does Java's Type Erasure accomplish when compiling generic code architectures?",
    "answers" : [ "It replaces type parameters with bounds or Object to ensure type safety without runtime overhead", "It converts object reference addresses into encrypted byte arrays", "It forces the compiler to duplicate class files for every unique type", "It restricts collection instances from scaling horizontally" ],
    "correctIndex" : 0
  }, {
    "question" : "Why are primitive types like int or char forbidden from being passed directly as parameters to Java Generics?",
    "answers" : [ "Database drivers cannot bind primitive types to statement query parameters", "Primitives consume too much L1 cache memory causing thread scheduler stalls", "The modularity guidelines written in module-info block primitive parameters", "Generics require object references on the heap; primitives do not inherit from Object" ],
    "correctIndex" : 3
  }, {
    "question" : "What happens if a catch block intercepts an exception and rethrows a new RuntimeException wrapper?",
    "answers" : [ "The current execution path breaks and the exception propagates up the thread stack frame", "The operating system kernel terminates the parent thread launcher instance", "The JVM defaults all text inputs back to empty strings", "The active relational database transaction commits its row status" ],
    "correctIndex" : 0
  }, {
    "question" : "What type of exception is a NullPointerException classified as inside the Java runtime architecture?",
    "answers" : [ "A Compile-time syntax flag anomaly", "An Unchecked Exception (RuntimeException)", "An Error exception condition", "A Checked Exception" ],

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"correctIndex" : 1
}, {
"question" : "What block in a try-catch-finally layout is guaranteed to execute regardless of
whether an exception is thrown?",
"answers" : [ "The throws block", "The try block", "The catch block", "The finally block" ],
"correctIndex" : 3
}, {
"question" : "How does the try-with-resources statement handle asset hygiene automatically on
block exits?",
"answers" : [ "It flushes the CPU cache logs to clear remaining variables", "It dispatches a
dedicated Garbage Collection sweep task", "It saves an execution checkpoint status state directly
back to tables", "It automatically invokes .close() on any resource object that implements
AutoCloseable" ],
"correctIndex" : 3
}, {
"question" : "What thread safety violation occurs when a thread caches local variable copies
instead of reading from main RAM?",
"answers" : [ "Illegal Thread State condition", "Race Condition anomaly", "Memory Visibility
Problem", "Deadlock layout lock" ],
"correctIndex" : 2
}, {
"question" : "Which thread-safe collection from java.util.concurrent should replace a standard
HashMap in high-concurrency environments?",
"answers" : [ "Collections.synchronizedMap", "Hashtable", "ConcurrentHashMap",
"LinkedHashMap module structure" ],
"correctIndex" : 2
}, {
"question" : "What modifier keyword forces Java threads to read and write variable values directly
from main memory?",
"answers" : [ "transient properties tracking", "synchronized", "volatile", "static" ],
"correctIndex" : 2
}]
}

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